

Focused Curation Report — *Pyricularia oryzae* PoMZ_10221 / F8U970

Hypothesis slug: rhamnose-enzyme-and-mat-regulatory-capacity **Focus** type: function_assignment **Gene:** PoMZ_10221 (UniProt F8U970, Swiss-Prot reviewed; EMBL JF740057 / AEH41994.1; genome QBZ54521.1) **Organism:** *Pyricularia oryzae* (= *Magnaporthe grisea/oryzae*), NCBITaxon:318829

Executive Judgment

Verdict on the seed hypothesis: REFUTED (for the MAT claims) / the UDP-rhamnose claim is SUPPORTED.

The seed proposes that F8U970 *retains* methionine adenosyltransferase (MAT) regulator activity, MAT-complex membership, and participation in S-adenosylmethionine (SAM) biosynthesis **in addition to** its experimentally supported UDP-rhamnose epimerase/reductase activity, and asks that each claim be adjudicated independently.

- The **UDP-rhamnose epimerase/reductase** function is **directly and experimentally supported** (IDA,

 22102281

This is the protein's primary, catalytically demonstrated function.

- The **three MAT-related claims** (MAT regulator activity, MAT complex, SAM biosynthesis) are **not supported by any direct, interaction, localization, or mutant evidence**. On F8U970 they exist **only as**  automated annotations propagated through PANTHER PTHR10491:SF4 ("Methionine adenosyltransferase 2 subunit beta"). They are best explained as **phylogenetic-propagation (paralog-lumping) over-annotations**, not a genuine second capacity.

Most important caveat: "Not proven" is not the same as "proven absent." No experiment has *tested* MAT-binding by F8U970. However, the burden of evidence, the twilight-zone homology to MAT2B (22%), the strong homology to true UDP-rhamnose synthases (63%), the absence of

any MAT/SAM mention in the primary paper, and the biological implausibility of a MAT2B-type beta regulatory subunit in fungi together make the "retained MAT capacity" claim unsupported for curation purposes.

The seed is also correct on one negative point: **this protein is not the fungal dihydroorotate dehydrogenase** claimed by an earlier automated report — UniProt and the primary literature unambiguously assign UDP-rhamnose synthesis.

Evidence Matrix

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
1	P 22102281 (Martinez et al., 2012)	Direct biochemical assay (NMR, enzymology)	Supports UDP-rhamnose function; Refutes MAT relevance (silent)	Is F8U970 a UDP-rhamnose epimerase/reductase?	"The second gene encodes a bifunctional UDP-4-keto-6-deoxyglucose-3,5-epimerase/-4-reductase that converts UDP-4-keto-6-deoxyglucose to UDP-rhamnose." Fungal pathway needs only two genes. No mention of MAT or SAM.	<i>Magnaporthe grisea</i> / <i>Botryotinia fuckeliana</i> vitro + expression
2	UniProt F8U970 (Swiss-Prot)	Database (curated)	Supports UDP-rhamnose; Qualifies MAT terms	Which GO terms are experimental vs computational?	UDP-rhamnose terms GO:0010489, GO:0010490, GO:0019300 are IDA ; MAT terms GO:0048269 (complex), GO:0048270 (regulator), GO:0006556 (SAM biosynthesis) are IEA:TreeGrafter only. Catalytic activities Rhea RHEA:56888 & RHEA:56892 (NADPH).	Reviewed entry
3	PANTHER PTHR10491 / SF4 (via UniProt DR)	Structural/ evolutionary (family model)	Competing / explains artifact	Source of MAT propagation	Family PTHR10491 = "dTDP-4-dehydrorhamnose reductase";	HMM/tree model

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					subfamily SF4 labeled " MAT2 subunit beta. " The family lumps UDP-/dTDP-sugar epimerase-reductases with the metazoan MAT2B regulator (shared NAD(P)-binding Rossmann/extended-SDR fold, IPR036291).	
4	This report — global NW alignment (BLOSUM62)	Computational (sequence)	Refutes MAT orthology; Supports UDP-rhamnose orthology	How close is F8U970 to MAT2B vs UDP-rhamnose synthases?	Identity to human MAT2B (Q9NZL9) = 22.0% (fold-level noise); to Arabidopsis UDP-L-rhamnose synthase RHM1 (Q9SYM5) C-domain = 62.8% . Yeast MAT (SAM2) 24.9% and human MAT2A 25.7% are alignment noise (different folds).	In silico
5	InterPro IPR005913 (RmlD), IPR029903, Pfam PF04321 (RmlD_sub_bind); UniProt SIMILARITY	Structural/ evolutionary	Supports UDP-rhamnose family assignment	What fold/ family?	F8U970 "belongs to the dTDP-4-dehydrorhamnose reductase family" (RmlD-like, NAD(P)-binding Rossmann). This is the sugar-nucleotide	Domain models

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
					reductase family, not a MAT catalytic fold.	
6	P 31496615 (Murray et al. 2019, review)	Review/ database	Qualifies / competing	Is the MAT2A/ MAT2B β - subunit system organism- general?	"Mammalian systems express two genes" for MATs; MATII = catalytic MAT2A + regulatory MAT2B β -subunit. The β -regulatory architecture is described in mammals, not fungi.	Human/ mammal liver
7	P 32046373 (An et al. 2020)	Mutant/ localization/ interaction	Qualifies / competing	What does MAT2B actually do?	MAT2B is a regulatory subunit that "interacts with G protein-coupled receptor kinase interacting ArfGAP1 [GIT1] to increase the activity of extracellular signal-regulated kinases (ERKs)"; nuclear/cytoplasmic in oocytes. Its function is protein-interaction/signaling scaffolding — a metazoan role with no counterpart evidenced for the fungal enzyme.	Mouse oocytes

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
8	UniProt G4NCL5 (this report)	Computational/ database (genome inventory)	Refutes SAM- biosynthesis claim for F8U970	Does F8U970 perform SAM biosynthesis in <i>P. oryzae</i> ?	<i>P. oryzae</i> encodes a dedicated canonical S- adenosylmethionine synthase G4NCL5 (EC 2.5.1.6; Pfam PF00438/PF02772/ PF02773; PANTHER PTHR11964) — a different gene and a different fold from F8U970 (PF04321, PTHR10491). SAM biosynthesis is handled by G4NCL5, not the UDP- rhamnose enzyme.	<i>P. oryzae</i> proteome
9	UniProt G4ML22 (this report)	Computational/ database (paralog)	Competing / explains artifact	Is the SF4 mislabel systematic?	A second <i>P. oryzae</i> RmlD-fold protein G4ML22 (Pfam PF04321) carries the identical PANTHER PTHR10491:SF4 "MAT2 subunit beta" assignment as F8U970. Multiple fungal sugar- nucleotide enzymes inherit the same spurious MAT label — direct evidence of systematic subfamily mis- annotation, not a genuine F8U970-	<i>P. oryzae</i> proteome

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
					specific second capacity.	

GO Curation Implications

Lead (requires curator verification):

- **RETAIN (experimental, core):**
- MF [GO:0010489](#) UDP-4-keto-6-deoxy-glucose-3,5-epimerase activity (IDA)
- MF [GO:0010490](#) UDP-4-keto-rhamnose-4-keto-reductase activity (IDA)
- BP [GO:0019300](#) rhamnose biosynthetic process (IDA)
- (Consider also a general [nucleotide-sugar biosynthetic process](#) / dTDP-4-dehydrorhamnose reductase family MF if the curation model prefers it; NADP binding CC/ MF is defensible from the reductase reaction.)
- **REMOVE / DO NOT PROPAGATE (over-annotation; [IEA:TreeGrafter](#) only):**
- CC [GO:0048269](#) methionine adenosyltransferase complex
- MF [GO:0048270](#) methionine adenosyltransferase regulator activity
- BP [GO:0006556](#) S-adenosylmethionine biosynthetic process

Rationale: no experimental/interaction/localization support; artifact of PTHR10491:SF4 lumping the fungal UDP-rhamnose enzyme with the divergent metazoan MAT2B regulator; biologically implausible in a fungus that lacks a MAT2B-type beta regulatory subunit. **Crucially, SAM biosynthesis in *P. oryzae* is performed by a separate dedicated canonical SAM synthase (G4NCL5, EC 2.5.1.6, PTHR11964), so GO:0006556 on F8U970 is redundant and incorrect.** A co-family paralog (G4ML22) carries the same spurious SF4 "MAT2 beta" tag, confirming the mislabel is systematic.

- **Do not recommend "protein binding"** — a specific, supported MF (the epimerase/reductase activities above) is available.

The evidence supports **MF + BP** (catalysis and rhamnose biosynthesis) as **core**, and argues the **MAT MF/BP/CC as non-core and removable/excludable** (candidate **NOT**/exclusion or simply do-not-propagate).

Mechanistic Scope

Immediate molecular function (direct): F8U970 catalyzes the committed epimerization + NADPH-dependent reduction step converting UDP-4-keto-6-deoxy-D-glucose → UDP-β-L-rhamnose (RHEA:56888 then RHEA:56892), the second enzyme of a two-gene fungal UDP-rhamnose pathway.

Downstream/context (not the tested molecular activity): rhamnose-containing glycans influence host–pathogen interactions (adhesion, recognition, virulence, biofilm). These are *pathway/phenotype consequences*, not molecular functions of this protein, and their tissue-specific regulation is a transcriptional observation.

The MAT "function" would require physical association with a MAT catalytic subunit and modulation of SAM synthase activity — none of which is demonstrated. It is inferred *only* from shared fold via automated propagation.

Conflicts and Alternatives

- **Paralog/family lumping (primary alternative explanation):** PANTHER PTHR10491 contains both sugar-nucleotide epimerase-reductases (Rm1D/RHM-type) and the metazoan MAT2B regulator, which share the extended-SDR / NAD(P)-binding Rossmann fold. TreeGraftor grafts F8U970 near a node carrying MAT2B-derived IBDs, exporting MAT terms onto a protein whose real neighbors are UDP-rhamnose synthases (63% identity to RHM1). The seed's own notes ("exact fungal accession absent from reference tree," "broad regulator IBD PTN000051877 not in the recovered graft path") are consistent with an unstable/over-reaching graft.
- **Organism-specific difference:** the source of the propagated IBDs, human MAT2B (Q9NZL9), is characterized specifically as the **mammalian** regulatory β-subunit of MATII

(**P** 31496615)

"Mammalian systems express two genes"), and its regulatory action operates through protein–protein interactions in metazoan signaling (binds GIT1, activates ERK;

P 32046373

Fungal methionine adenosyltransferases (e.g., *S. cerevisiae* Sam1/Sam2) are single-subunit homo-oligomeric SAM synthetases without a MAT2B-type β regulatory subunit, so there is no $\alpha 2\beta 2$ MAT complex for F8U970 to join in *P. oryzae*. (Stated as inference from the mammalian-restricted MAT2B literature; not directly disproven experimentally.)

- **Database carry-over:** a prior automated ("OpenAI") report mislabeled the protein as dihydroorotate dehydrogenase; that is a separate, clearly incorrect assignment and should not influence curation.
- **No competing experimental evidence** supports MAT function.

Knowledge Gaps

1. **Direct test of MAT association.** Checked: no interaction/localization data in UniProt or the primary paper. Matters because the seed asks specifically about "catalytic/regulatory coexistence." Resolved by: co-IP / pull-down of F8U970 with *P. oryzae* SAM synthetase, or AlphaFold-Multimer interface scoring — neither currently in evidence.
2. **Exact TreeGrafter version/graft path.** Checked: UniProt reports the SF4 assignment and IEA terms but not the frozen graft node internals; I could not programmatically resolve the specific PTN008946252 vs PTN000051875/PTN000051877 IBD propagation logic. Matters for reproducing exactly which IBD exported the MAT terms. Resolved by: inspecting the PAINT/TreeGrafter annotation for PTHR10491 at the frozen version.
3. **NADP cofactor CC/MF specificity.** The reductase uses NADPH (RHEA:56892) but an explicit NADP binding MF is not asserted; low-priority, easily added.

Discriminating Tests

1. **AlphaFold-Multimer / co-IP:** test whether F8U970 forms a complex with the *P. oryzae* methionine adenosyltransferase catalytic subunit. Predicted result if hypothesis false: no stable interface (unlike human MAT2A–MAT2B).

2. **Reciprocal-best-hit / phylogeny with proper outgroups:** place F8U970 among fungal/plant UDP-rhamnose synthases vs MAT2B clade; expect robust grouping with RHM-type enzymes (consistent with 63% identity), not MAT2B.
3. **Δgene metabolite phenotype:** knockout should reduce UDP-rhamnose / rhamnoglycans (already implied by pathway), and should **not** perturb SAM pools if MAT function is absent.
4. **Active-site residue audit:** confirm RmlD reductase catalytic/NADP-binding residues are intact (they are, given retained catalysis), whereas MAT2B-specific regulatory-interface residues are absent.

Curation Leads (verify before applying)

- **Action change:** downgrade/remove the MAT-related `IEA:TreeGrafter` terms (GO:0048269, GO:0048270, GO:0006556) as paralog-driven over-annotations; retain the IDA UDP-rhamnose terms as core.

- **Candidate reference to cite for core function:**

P 22102281

— verify snippet: *"The second gene encodes a bifunctional UDP-4-keto-6-deoxyglucose-3,5-epimerase/-4-reductase that converts UDP-4-keto-6-deoxyglucose to UDP-rhamnose."*

- **Candidate GO (retain/core):** GO:0010489, GO:0010490 (MF); GO:0019300 (BP). Optionally add `NADP binding`.
- **Candidate GO (exclude/non-core):** GO:0048269, GO:0048270, GO:0006556.
- **Suggested curator question:** Is there *any* organism-specific evidence (interaction, complexome, localization) that a MAT2B-type beta regulator exists in *P. oryzae*? If not, the CC term is untenable.
- **Suggested experiment:** AlphaFold-Multimer/co-IP of F8U970 with *P. oryzae* SAM synthetase to formally close the "coexistence" question.
- **Note:** flag and discard the earlier "dihydroorotate dehydrogenase" automated assignment as incorrect.

Provenance

- Primary literature:

 22102281

abstract (PubMed).

- Records: UniProt Swiss-Prot F8U970 JSON (GO evidence codes, catalytic activities, PANTHER/InterPro/Pfam DR lines).
- Computed: global Needleman–Wunsch (BLOSUM62) pairwise identities — F8U970 vs Q9NZL9 (22.0%), P31153 (25.7%), P19358 (24.9%), Q9SYM5 (62.8%). Executed code and outputs are retained in this iteration's execution log.
- Computed (Iteration 3): UniProt proteome query of *P. oryzae* (taxon 318829) — the organism's canonical SAM synthase is **G4NCL5** (EC 2.5.1.6, Pfam PF00438/PF02772/PF02773, PTHR11964), distinct from F8U970; RmlD-fold paralog **G4ML22** (PF04321) shares the same spurious PTHR10491:SF4 "MAT2 beta" label. Executed code and outputs retained in the Iteration 3 execution log.

Decision Table (leads — verify before applying)

GO ID	Aspect	Term	Current evidence	Lead action
GO:0010489	MF	UDP-4-keto-6-deoxy-glucose-3,5-epimerase activity	IDA ()	Retain (core)
GO:0010490	MF	UDP-4-keto-rhamnose-4-keto-reductase activity	IDA ()	Retain (core)
GO:0019300	BP	rhamnose biosynthetic process	IDA ()	Retain (core)
GO:0048270	MF	methionine adenosyltransferase regulator activity	IEA:TreeGrafter only	Remove / do-not-propagate (over-annotation)
GO:0048269	CC	methionine adenosyltransferase complex	IEA:TreeGrafter only	Remove / do-not-propagate (over-annotation)
GO:0006556	BP	S-adenosylmethionine biosynthetic process	IEA:TreeGrafter only	Remove (redundant; SAM synthesis done by G4NCL5)
(optional)	MF	NADP binding	inferable from RHEA:56892 reductase	Consider adding

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